NAME

mbmapscale − Outputs the local scaling between geographic coordinates and meters east and north at a specified latitude.

VERSION

Version 5.0

SYNOPSIS

mbmapscale [**−L***latitude* **−A −V −H]**

DESCRIPTION

The program **mbmapscale** outputs the scaling between geographic coordinates (longitude and latitude) and local meters east and north at a user defined latitude. This is useful for converting between degrees and meters when working with local Cartesian approximations of swath data positions.

By default the scaling is calculated using the WGS72 ellipsoid. If the **−A** flag is given, the scaling is instead calculated using the 1866 Clark Spheroid as employed by the AlvinXY local coordinate convention.

The map scale is written to stdout in the form of meters per degree longitude and latitude, and the inverse (degrees per meter, i.e. "meters to degree").

MB-SYSTEM AUTHORSHIP

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OPTIONS

−H This "help" flag causes the program to print out a description of its operation and then exit immediately.

−A Calculate the map scale using the 1866 Clark Spheroid as employed by the AlvinXY local coordinate convention, rather than the default WGS72 ellipsoid.

−L *latitude*
Sets the latitude (in decimal degrees, positive north) at which the map scale is calculated. If **−L** is not specified, the scale is calculated at the equator (latitude 0.0).

−V Normally, **mbmapscale** only prints out the calculated scale values. If the **−V** flag is given, then **mbmapscale** works in a "verbose" mode and also outputs the program version and control parameters being used.

EXAMPLES

Suppose one wishes to obtain the map scale at 36.8 degrees north latitude using the default WGS72 ellipsoid. The following command:

```
mbmapscale −L36.8
```

yields a result of the form:

Local scaling between degrees longitude and latitude and meters east and north:

Using WGS72 ellipsoid
Meters per degree longitude: 89244.230
Meters per degree latitude: 110973.870
Meters to degree longitude: 0.000011205
Meters to degree latitude: 0.000009011

Suppose instead one wishes to obtain the map scale at the same latitude using the AlvinXY (1866 Clark Spheroid) convention:

mbmapscale −L36.8 −A

SEE ALSO

mbsystem(1), **mbtime**(1), **mbio**(1)

BUGS

Never bugs, but perhaps our good friends, the terrestrial isopods.